

Account

1. Feature Overview

Open accounts under a customer, enforce account policies (opening deposit / min balance metadata), search accounts, update status, and post simple deposits that credit balances.

Status: Implemented for open / search / status / deposit / policies. **No** transaction ledger, withdraw, or transfer.

2. Business Purpose

Hold customer balances and product types (SAVINGS, CURRENT, SALARY, FD, RD, LOAN) with branch-coded account numbers.

3. User Workflow

- 1. `/app/accounts` — search list, Deposit action
- 2. Customer detail — Add Current (loads policy → opening deposit dialog if required) / Add Loan
- 3. Customer create — optional first account
- 4. Policies seeded at startup; create policy via API

4. Execution Flow

See [CALL_FLOW.md](#), [deposit](#), [get active account policy](#).

5. Database Tables

Table	Purpose
<code>account</code>	Balances, status, number parts
<code>account_policy</code>	Per type+currency rules
<code>account_ordinal_seq</code>	Ordinal for number generation
<code>customers</code>	Owner FK

6. REST APIs

Method	Path	Roles
POST	/accounts	ADMIN, EMPLOYEE
GET	/accounts	ADMIN, EMPLOYEE, MANAGER
GET	/accounts/customer/{customerId}	ADMIN, EMPLOYEE, MANAGER
PUT	/accounts/{accountId}/status	ADMIN, EMPLOYEE
POST	/accounts/{accountId}/deposit	ADMIN, EMPLOYEE
POST	/account-policies	Authenticated*
GET	/account-policies?accountType¤cyCode	Authenticated*

*Restrict to ADMIN — deferred hardening.

7. Controllers

- AccountController
- AccountPolicyController

8. Services

- AccountService / AccountServiceImpl — openAccount, getAccountsByCustomerId, updateAccountStatus, deposit, searchAccounts
- AccountPolicyService / AccountPolicyServiceImpl — createPolicy, getActivePolicy
- AccountPolicyInitializer — seed INR policies

9. Repositories

- AccountRepository (+ getNextOrdinal() native query)
- AccountPolicyRepository

10. DTOs

OpenAccountRequest, AccountResponse, UpdateAccountStatusRequest, DepositRequest, CreateAccountPolicyRequest, AccountPolicyResponse

11. Entities / Enums

- Account, AccountPolicy
- AccountType, AccountStatus, CurrencyCode

12. Utility Classes

- AccountNumberGenerator.generate(...)

- `AccountSpecification.matching(...)`

13. Configuration Classes

`SecurityConfig` matches for `/accounts/**`; policy paths fall under `anyRequest().authenticated()`.

Frontend: `core/services/account.ts`, `account-policy.ts`, `features/accounts/`, `features/opening-deposit-dialog/`

14. Validation Rules

- Open: customer exists; active policy in effective window; if `openingDepositRequired` then amount $\geq$ `requiredOpeningDeposit`
- Deposit: account ACTIVE; amount positive
- Policy create: Bean Validation on `CreateAccountPolicyRequest`

15. Security Rules

Account mutations ADMIN/EMPLOYEE; reads include MANAGER. Policy endpoints currently any authenticated user.

16. Exception Handling

Business exceptions for missing account/customer/policy and validation failures $\rightarrow$ `GlobalExceptionHandler`.

17. Logging

SQL debug in dev; no structured account event log.

18. Audit Events

Account has `created_by` / timestamps; `AccountPolicy` uses `AuditableEntity`. No immutable audit trail of deposits.

19. Testing Strategy

- Open CURRENT with deposit $< 5000 \rightarrow$ reject
- Open CURRENT with $\geq$ policy min $\rightarrow$ balances equal deposit
- Deposit on ACTIVE $\rightarrow$ balances and `creditCount` increase
- Deposit on non-ACTIVE $\rightarrow$ reject
- GET policy CURRENT/INR returns seed row
- UI: Add Current opens dialog when required

20. Future Extension Guide

- Persist ledger rows on every credit/debit
- Withdraw / transfer with limits
- Account detail page
- Status transition state machine

- Seed LOAN policy or block LOAN until product module exists
- Maker-checker for large deposits

Future Modification Guide

Requirement: Increase required opening deposit for CURRENT/INR

Item	Detail
Prefer	Update/insert <code>account_policy</code> row (API <code>POST /account-policies</code> or DB) — not hardcode UI
Files	<code>AccountPolicyInitializer.java</code> (seed), or runtime policy via <code>AccountPolicyServiceImpl.createPolicy</code>
Classes	<code>AccountPolicy</code> , <code>AccountPolicyServiceImpl</code>
Methods	<code>createPolicy()</code> , <code>getActivePolicy()</code> ; enforced in <code>AccountServiceImpl.openAccount()</code>
Impact	<code>OpeningDepositDialog</code> min; customer onboarding amounts

Requirement: Change deposit to write a ledger entry

Item	Detail
Files	New Transaction module + <code>AccountServiceImpl.deposit</code> / <code>openAccount</code>
Methods	<code>deposit()</code> must call transaction create after balance update (or same TX)
Impact	Dashboard <code>todayTransactionCount</code> ; Reports; Audit

Requirement: Change account number format

Item	Detail
Files	<code>AccountNumberGenerator.java</code> , possibly <code>AccountServiceImpl.openAccount</code>
Methods	<code>generate(branchCode, accountType, ordinal, currency, checkDigit)</code>
Impact	Display/search only for new accounts; old numbers unchanged

Requirement: Allow deposit on FROZEN accounts

Item	Detail
Files	<code>AccountServiceImpl.java</code>
Methods	<code>deposit()</code> status check
Impact	Status rules doc; UI Deposit button enablement

Call hierarchy (open account)

```
AccountController.openAccount()  
    ↓  
AccountServiceImpl.openAccount()  
    ↓  
CustomerRepository.findById()  
    ↓  
AccountPolicyRepository.findByAccountTypeAndCurrencyCodeAndActiveTrue()  
    ↓  
AccountRepository.getNextOrdinal()  
    ↓  
AccountNumberGenerator.generate()  
    ↓  
AccountRepository.save()
```

Call hierarchy (deposit)

```
AccountController.deposit()  
    ↓  
AccountServiceImpl.deposit()  
    ↓  
AccountRepository.findById()  
    ↓  
AccountRepository.save()
```